

AERO50005 TUTORIAL 1

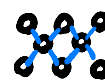
Basics

Questions

Question 1. Using sketches where appropriate, define the following:

(a) Polymorphism *pure metal have more than one crystal structures at different temperature*

(b) Substitutional solid solution with ordering



*strength ↑
ductility ↓*

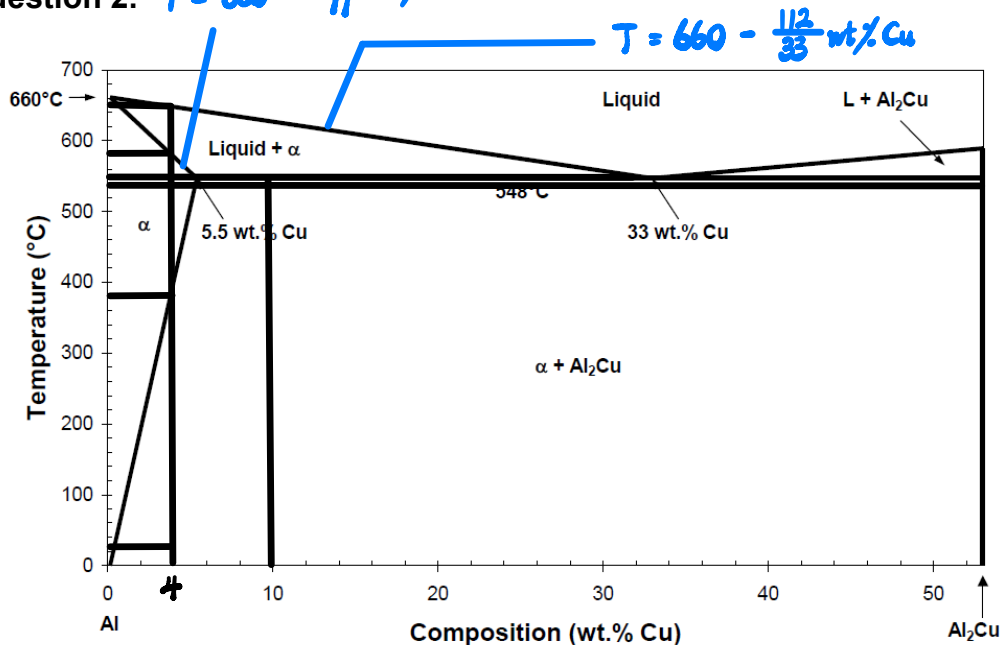


*distort the
dislocation field*

(c) Four types of interphase boundaries

in order from small to big $\delta\sigma$: coherent, coherency strain, semi-coherent, incoherent

Question 2. $T = 660 - \frac{224}{11} \text{ wt\% Cu}$



The phase diagram above is the aluminium rich end of the Al-Cu system. This is an important alloy system as many aerospace and high strength aluminium alloys are based on this system.

(a) Referring to this phase diagram answer the following questions for an Al-4 wt.% Cu alloy being cooled from 700°C down to room temperature.

(i) By calculation,

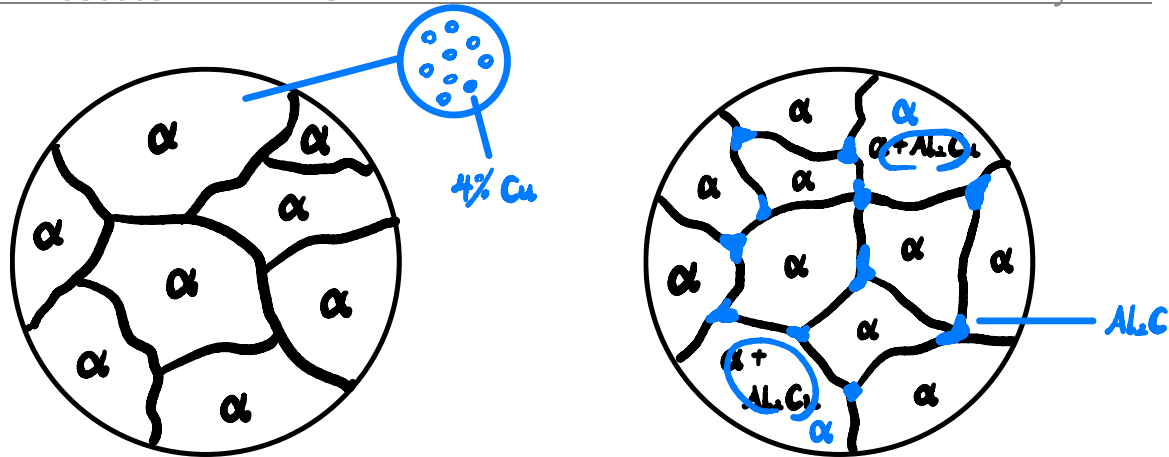
at what temperature will this alloy start to solidify 646.424 °C;

at what temperature will this alloy end solidification 580 °C

(ii) In the circle below sketch and label the microstructure you would expect to see at 550°C and at room temperature.

At 550°C:

At room temperature:



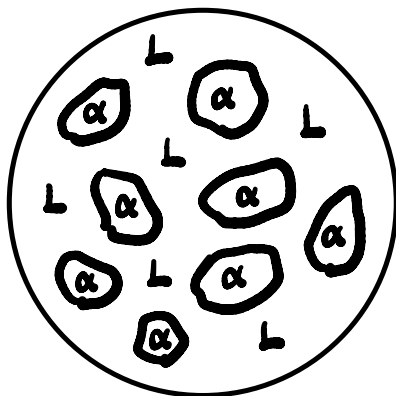
(iii) What are the composition and proportion of phases present at room temperature? $f_{\alpha} + f_{Al_2Cu} = 1$, $0.5\% \text{ wt } f_{\alpha} + 53\% \text{ wt } f_{Al_2Cu} = 4\% \text{ wt}$

$$\alpha + Al_2Cu : f_{\alpha} = \frac{4}{15} = 26.67\%, f_{Al_2Cu} = \frac{11}{15} = 73.33\%$$

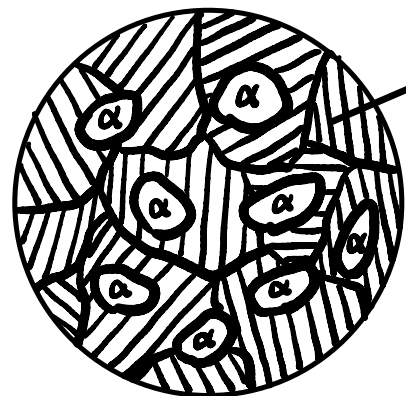
(b) Referring to the phase diagram answer the following questions for an Al-10 wt.% Cu alloy being cooled from 700°C down to room temperature.

(i) In the circle below sketch and label the microstructure you would expect to see at 550°C and at room temperature.

At 550°C:



At 540°C:



(ii) What are the composition and proportion of phases present at 550°C.

$$f_{\alpha} + f_L = 1, 5.5\% \text{ wt } f_{\alpha} + 33\% \text{ wt } f_L = 10\% \text{ wt} \quad \therefore \alpha + L : f_{\alpha} = \frac{16}{35} = 45.7\%, f_L = \frac{9}{35} = 25.7\%$$

(iii) What are the composition and proportion of phases present at 540°C. [SEP]

$$f_{\alpha} + f_{Al_2Cu} = 1, 5.5\% \text{ wt } f_{\alpha} + 53\% \text{ wt } f_{Al_2Cu} = 10\% \text{ wt} \quad \therefore \alpha + L : f_{\alpha} = \frac{26}{35} = 74.3\%, f_{Al_2Cu} = \frac{9}{35} = 25.7\%$$

(iv) What is the ratio of primary alpha (alpha formed before the eutectic reaction takes place) to eutectic microstructure at 540°C.

$$\text{ratio of primary } \alpha = \frac{74.3\%}{25.7\%} = 2.89$$

Question 3.

(a) What is the change in interfacial area when a sphere of radius 1mm merges with a sphere of radius 2mm to create a single, larger sphere?

$$V_{\text{final}} = \frac{4}{3}\pi[(1)^3 + (2)^3] = 12\pi (\text{mm}^3) \quad \therefore r_f = 2.08 (\text{mm})$$

$$\therefore \Delta A = 4\pi[r_f^2 - (1^2 + 2^2)] = -8.46 (\text{mm}^2)$$

(b) In your kitchen, mix cooking oil and vinegar. Stir the mixture well and leave to rest. Describe what happens and explain why.

Some oil will refuse to mix with vinegar.

This is because solubility limit has been reached, second phase will be formed.

Question 4.

Compare and contrast displacive and diffusive phase transformations.

Displacive phase transformations required fast cool, form a metastable martensite which distort lattice, make material very hard, only happened on polymorphism.

Diffuse phase transformation happened on slow cool, where the mass transport by atomic motion, form a stable nucleation. Solutions provide more details

Question 5.

Explain why the bubbles in your beer form at the edge of the glass?

This is very similar to the heterogeneous nucleation. Bubble more likely form at the edge of glass as glass-edge is an exist interfacial area. energy required to form bubble is reduced.

Question 3.

(a) What is the change in interfacial area when a sphere of radius 1mm merges with a sphere of radius 2mm to create a single, larger sphere?

(b) In your kitchen, mix cooking oil and vinegar. Stir the mixture well and leave to rest. Describe what happens and explain why.

Many small droplets of oil. these droplets may contact then merge to form fewer and larger droplets. to reduce interfacial area the minimise interfacial energy.

Question 4.

Compare and contrast displacive and diffusive phase transformations.

Question 5.

Explain why the bubbles in your beer form at the edge of the glass?